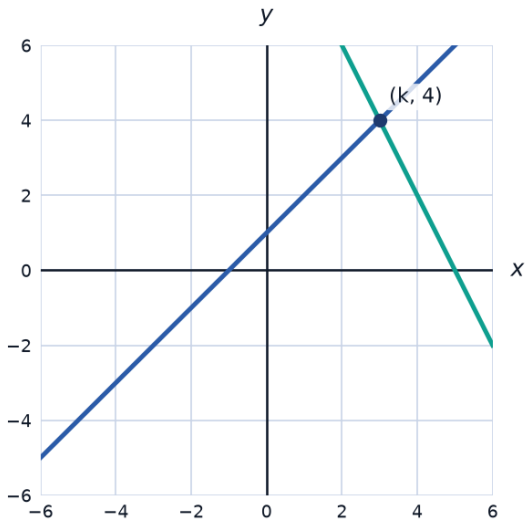
Linear Equations & Lines

ClipSAT Printable Practice Suite (clipsat.org) • Topic 1.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

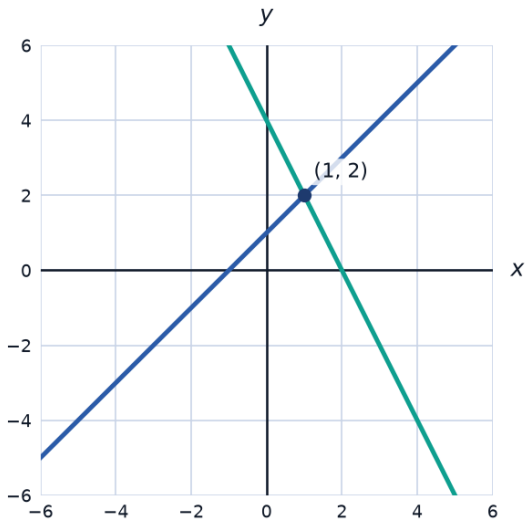
1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the system of linear inequalities represented by the shaded region in the graph below. The boundary lines are given by $y = -2x + 10$ and $y = x + 1$. The point $(k, 4)$ lies on the boundary line $y = -2x + 10$ and is within the feasible region. Determine the value of k.



2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the system of linear equations graphed below: $\{2x + y = 4; x - y = -1$ What is the solution (x, y) to this system?



3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the system of linear equations: $\{2x + y = 7; x - y = 2$ What is the value of x?

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4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the system of linear equations: $\{2x + y = 5; x - y = 1$ What is the solution (x, y) for this system?

5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Let the rational function be defined as $f(x) = \frac{x^2 - ax + 6}{x^2 - 5x + 6}$. If the graph of $y = f(x)$ has a removable discontinuity (a hole) at $x = 2$, determine the value of the parameter a.

6 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the function $f(x) = x^2 - 4x + 7$. The graph of $y = f(x)$ is subjected to a sequence of transformations. First, it is translated by a units horizontally and b units vertically. Let this transformed function be $g(x)$. Subsequently, the graph of $y = g(x)$ is reflected in the x-axis to produce the graph of $y = h(x)$. If the vertex of the graph of $y = h(x)$ is at $(-2, -3)$, determine the value of $a^2 + b^2$.

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The graph of the linear equation $y = mx + c$ is shown below. The line passes through the point $(-3, -11)$ and intersects the y-axis at $(0, k)$. If the slope of the line is -3, determine the value of k.

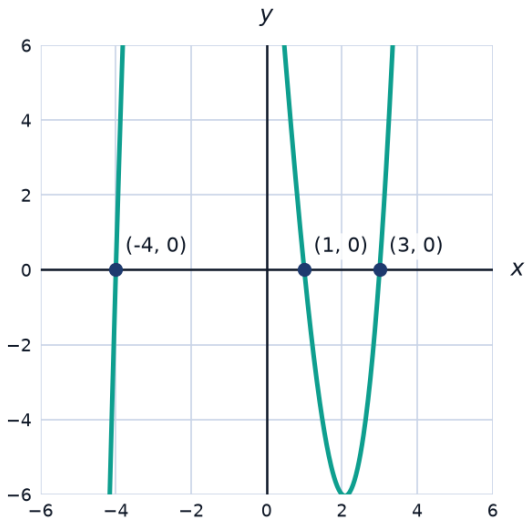
Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

What is the solution (x, y) to the system of equations below? $x + y, = 8; 2x - y, = 7$

9 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

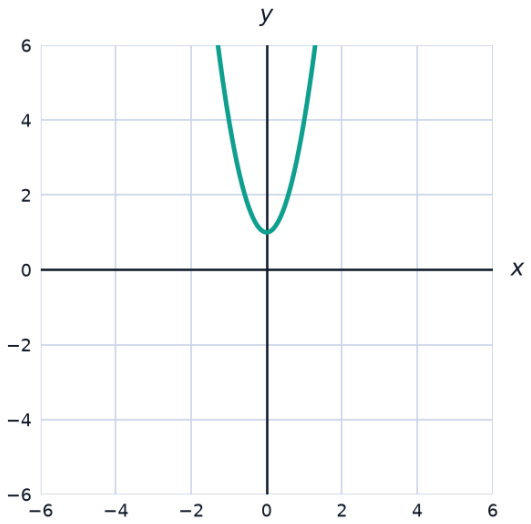
Consider the polynomial $P(x) = x^3 - 13x + 12$. The graph of $y = P(x)$ is shown below. The polynomial can be factored into the form $P(x) = (x - a)(x - b)(x - c)$ where $a < b < c$. Determine the value of $c - a$.



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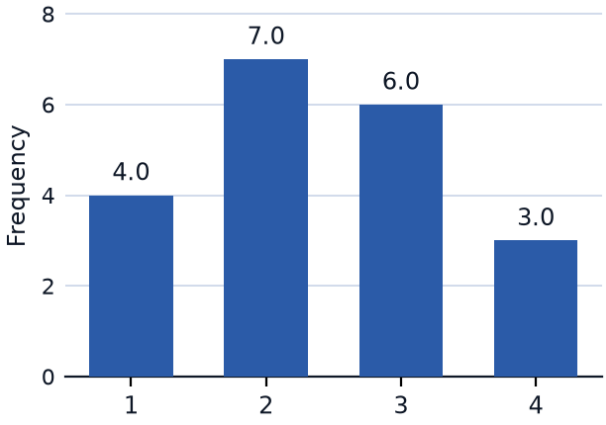
10 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The graph of the function $f(x) = ax^2 + b$ is shown below. The graph passes through the points $(-2, 13)$ and $(3, 28)$. Determine the value of b.



11 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

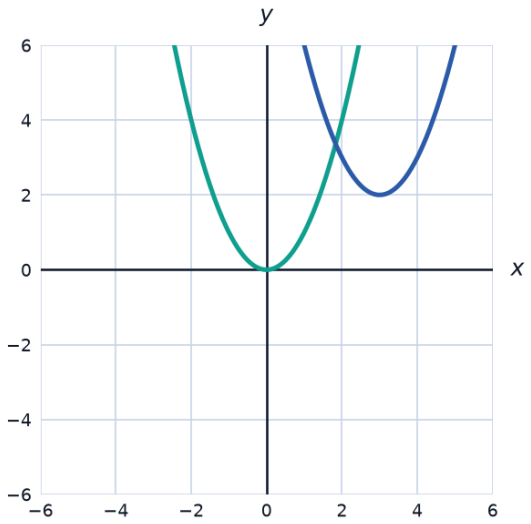
The bar graph shows the frequency distribution of scores obtained by 20 students on a 5-point quiz, where no student originally scored 5. A group of k additional students take the quiz later, and each of these students earns a score of 5. If the mean score of the combined group of $20 + k$ students is equal to 3, what is the value of k?



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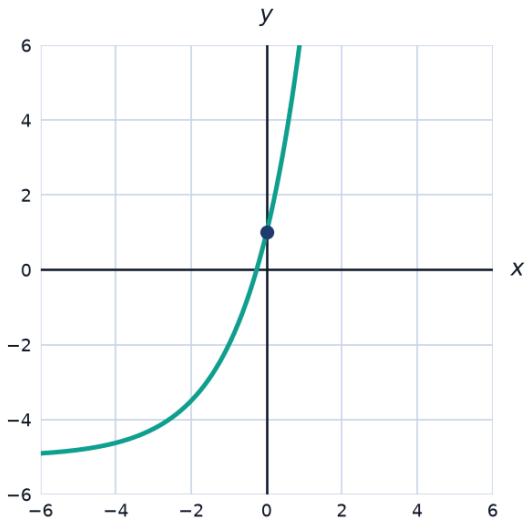
12 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The graph of the function $f(x) = x^2$ is shown below. It is transformed to the graph of $g(x) = (x - 3)^2 + 2$. Which transformation maps $f(x)$ to $g(x)$?



13 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

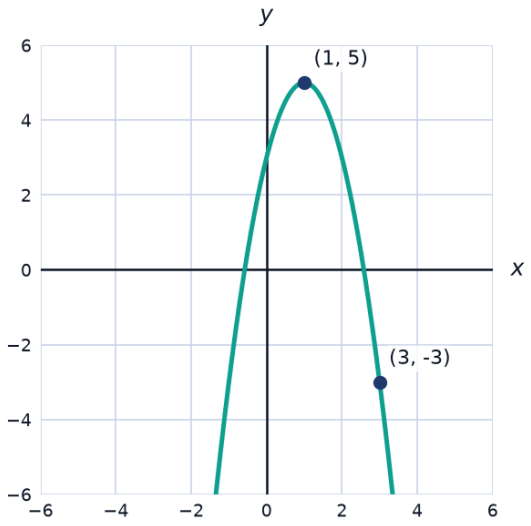
The graph of the function $f(x) = a \cdot b^x + c$ is shown below. The graph passes through the points $(0, 1)$ and $(1, 7)$, and has a horizontal asymptote at $y = -5$. Determine the values of a and b.



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14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

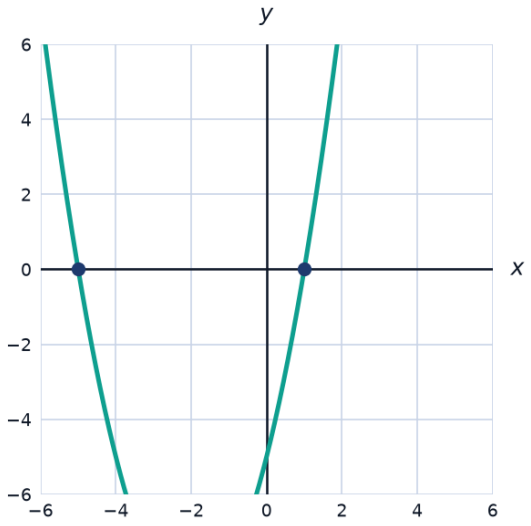
The graph of the quadratic function $f(x) = ax^2 + bx + c$ is shown below. The vertex of the parabola is at $(1, 5)$ and the parabola passes through the point $(3, -3)$. Determine the value of a.



Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Functions & Transformations

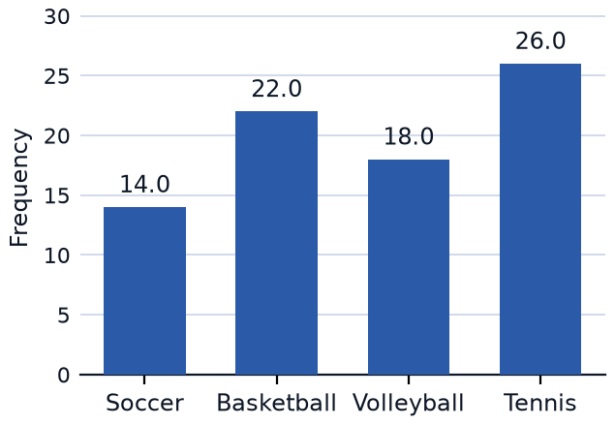
The graph of the quadratic function $f(x) = ax^2 + bx - 5$ is shown below. The function has roots at $x = -5$ and $x = 1$. Find the value of a.



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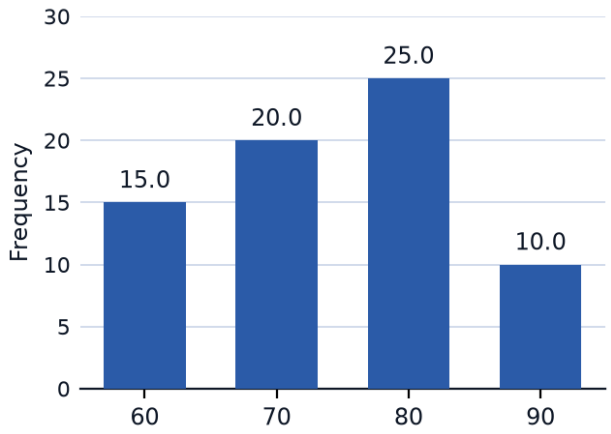
16 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

The bar chart below shows the number of students who participated in different sports at a school. What is the total number of students who participated in basketball and volleyball?



17 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.01 Systems & Inequalities

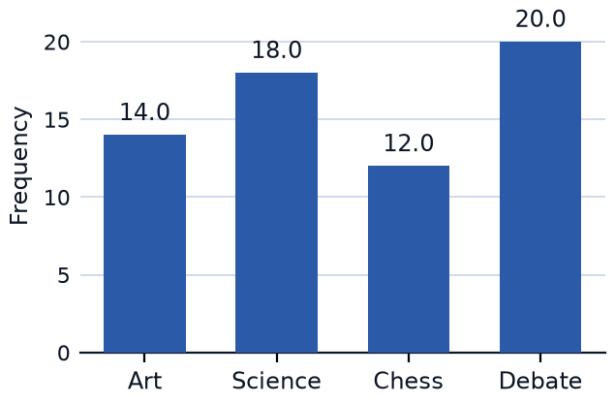
The following bar chart shows the number of students who achieved different scores on a recent quiz. What is the mode of the quiz scores?



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18 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 7.01 Sequences & Series

The bar chart below shows the number of students in four different clubs at a school. What is the total number of students in the Science Club and the Chess Club combined?



19 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 8.01 Factoring Polynomials

Let S be the set of all distinct real solutions to the equation $(x^2 - x + 1)^4 - 10x^2(x^2 - x + 1)^2 + 9x^4 = 0$. What is the sum of the squares of the elements of S?

20 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 9.01 Radicals & Exponents

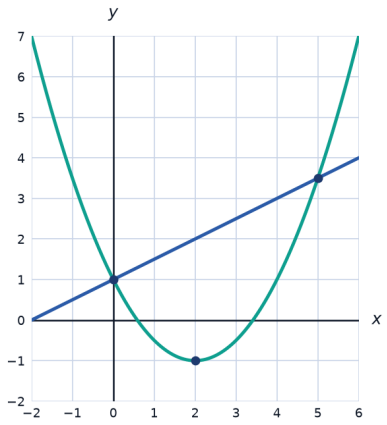
The rational function $R(x) = \frac{x^2 - 7x + k}{x - 3}$ has a removable discontinuity (a hole) at $x = 3$. For what value of k does this occur?

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 2x + 3$ in review: the real number system, evaluate $f(2)$.



a. 7

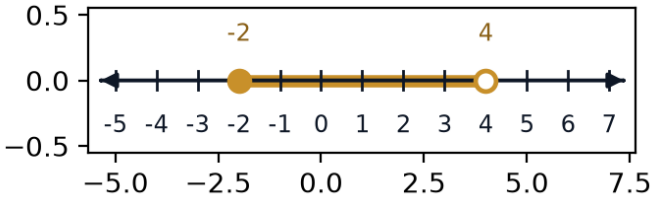
b. 9

c. 4

d. 11

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.



a. $x = 2$ and $x = 5$

b. $x = 2$ and $x = 4$

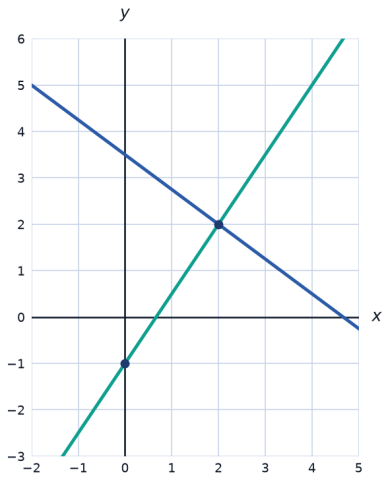
c. $x = -2$ and $x = -4$

d. $x = 6$ and $x = 8$

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 3x + 4$ in review: integer operations, evaluate $f(2)$.



a. 10

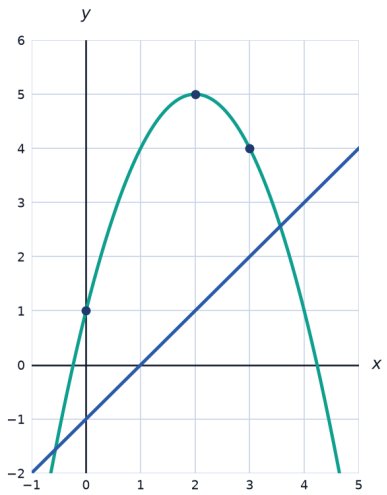
b. 13

c. 6

d. 14

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 4x + 5$ in review: operations with rational numbers, evaluate $f(2)$.



a. 13

b. 17

c. 8

d. 17

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25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for review: the real number system consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean?

a. 14.5

b. 15.5

c. 16.5

d. 18.5

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for review: integer operations consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5

d. 19.5

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for review: exponents and radicals consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5

d. 21.5

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 5x + 6$ in review: exponents and radicals, evaluate $f(4)$.

a. 26

b. 31

c. 20

d. 30

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for review: laws of exponents consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5

d. 22.5

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 6x + 7$ in review: laws of exponents, evaluate $f(4)$.

a. 31

b. 37

c. 24

d. 35

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 2x + 8$ in units resulting from a formula or graph, evaluate $f(3)$.

a. 14

b. 16

c. 6

d. 18

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

In the xy-plane, line ℓ_1 has the equation $3x + 4y = 24$. Line ℓ_2 is perpendicular to ℓ_1 and intersects ℓ_1 at a point P in the first quadrant. If line ℓ_1, line ℓ_2, and the x-axis enclose a triangle with an area of 6, what is the x-coordinate of P?

a. $\frac{12}{5}$

b. 3

c. $\frac{24}{5}$

d. $\frac{32}{5}$

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider a triangle ABC. The line segment AD is the angle bisector of angle BAC, where D lies on BC. Points E and F are on AC and AB respectively, such that EF is parallel to BC. Given that AB = 10 units, AC = 15 units, BD = 6 units, and the area of triangle AEF is $\frac{4}{9}$ of the area of triangle ABC. Determine the length of CD in units.

a. 7.5

b. 9

c. 12

d. 15

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34 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

A farmer is planning to fence a rectangular plot of land adjacent to a straight river. The farmer has 100 meters of fencing material. The side of the plot along the river does not require fencing. Let the dimensions of the rectangular plot be x meters (perpendicular to the river) and y meters (parallel to the river). The total area of the plot is given by $A = xy$. If the farmer wants to maximize the area, what are the dimensions x and y that achieve this maximum area?

a. $x = 25$ m, $y = 50$ m

b. $x = 30$ m, $y = 40$ m

c. $x = 20$ m, $y = 60$ m

d. $x = 35$ m, $y = 30$ m

35 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

In the xy-plane, line ℓ passes through the points $(-2, 5)$ and $(4, -1)$. Line k is perpendicular to line ℓ and has the equation $ax + 2y = c$, where a and c are constants. If line k passes through the point $(3, 1)$, what is the value of c?

a. -4

b. 2

c. 4

d. 8

36 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

Consider a line that passes through the point $(2, 5)$ and has a slope of -3. If the equation of this line is written in the form $y = mx + c$, and it is also known that the point $(k, -7)$ lies on this line, determine the value of k.

a. k = 5

b. k = 6

c. k = 7

d. k = 8

37 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Consider a line passing through the points $A(2, k)$ and $B(6, 10)$. If the slope of the line segment AB is m and $m = \frac{1}{2}$, find the value of k.

a. $k = 6$

b. $k = 8$

c. $k = 12$

d. $k = 4$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 7.01 Sequences & Series

In the xy-plane, the graph of the linear equation $4x - 7y = 28$ intersects the x-axis at the point $(c, 0)$. What is the value of c?

a. 7

b. -4

c. 4

d. 28

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 8.01 Factoring Polynomials

A line passes through the points $(2, 3)$ and $(6, 11)$. What is the slope of this line?

a. 2

b. $\frac{1}{2}$

c. -2

d. $-\frac{1}{2}$

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 9.01 Radicals & Exponents

Consider the line with the equation $y = 2x + 3$. What is the y-intercept of this line?

a. 3

b. 2

c. $-3/2$

d. $-2/3$

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 10.01 Rational Expressions

A line has slope -2 and passes through $(-6, 3)$. What is its y-intercept?

a. -11

b. -9

c. 9

d. -8

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 11.01 Statistics & Data Analysis

What is the slope of the line through $(1, -1)$ and $(3, 7)$?

a. 3

b. 4

c. $\frac{1}{4}$

d. -4

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 2.01 Quadratic Functions

Consider the system of linear equations: $kx + 12y = 2k + 6\ 3x + ky = 9$ where k is a constant. For which value of k does the system have no solution?

a. -6

b. -2

c. 2

d. 6

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Exponents & Polynomials

Consider the system of linear equations: x Which of the following points is the solution to this system?

a. $(1, 3)$

b. $(2, 5)$

c. $(3, 7)$

d. $(0, 1)$

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Functions & Transformations

How many solutions does a consistent independent system of two linear equations have?

a. Exactly one

b. None

c. Two

d. Infinitely many

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